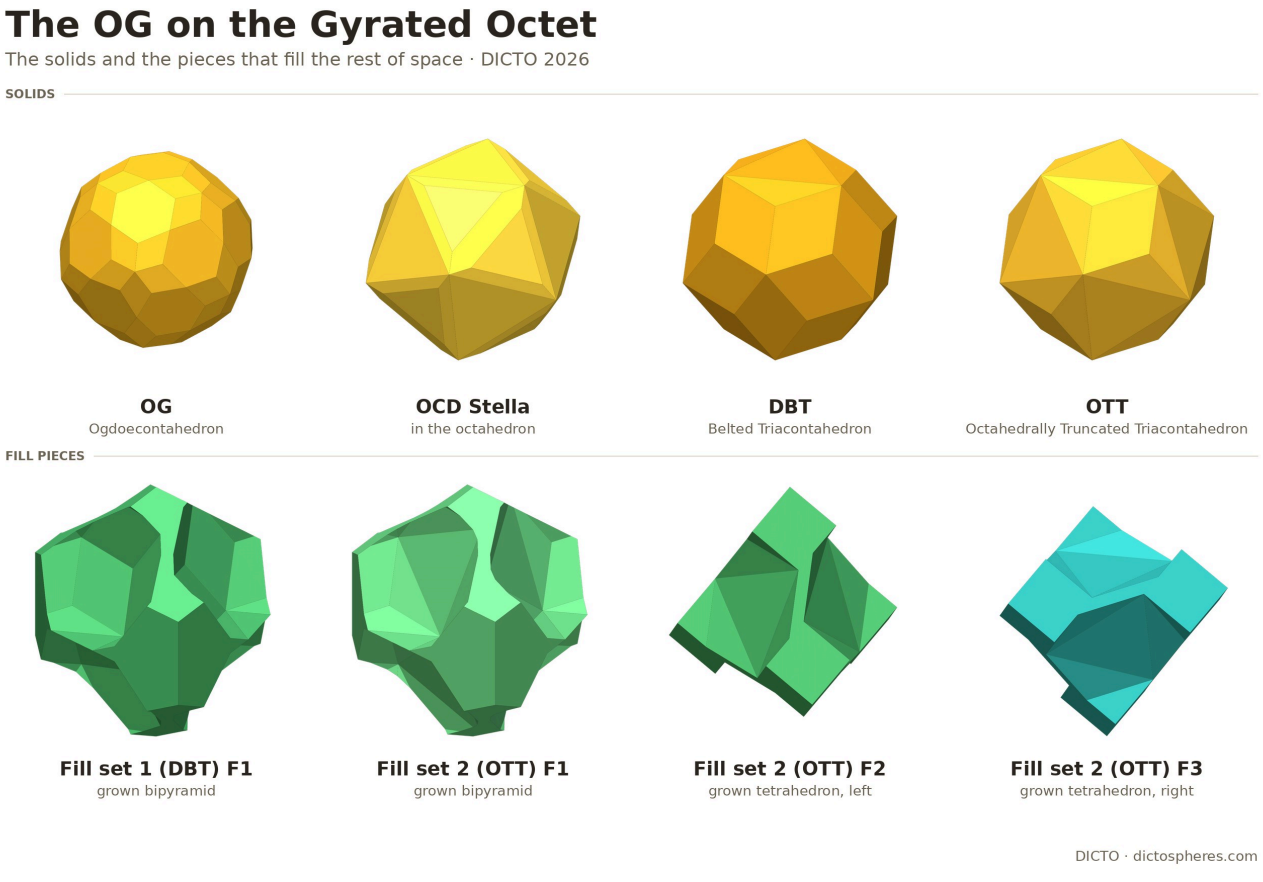
The DICTO Ogdoecontahedron on the Gyrated Octet

Twins at every layer: the Ogdoecontahedron, its OCD Stella, the DICTO Belted and the DICTO Octahedrally Truncated Triacontahedra in the gyrated tetrahedral–octahedral honeycomb, with one cell kit for every stacking and two-shape fill sets

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Version 5 · 5 November 2026 (adds a pointer to the cut-OG kits of Pentagons for the Kites; Version 4 clarified that it is the DICTO Jewel's packing, not the solid, that does not twin; Version 3 added the DICTO OTT, the DICTO Fill sets and the Penrose-core boundary pieces, and showed the DBT as a solid of its own) · DICTO DOI: 10.5281/zenodo.23484012 · Companion to The DICTO Ogdoecontahedron on the Octet Truss (DICTO DOI: 10.5281/zenodo.23457016) · Licence CC BY-SA 4.0 · Scripts, exact checks and data: https://github.com/DICTOR-Master/kaleidohedra/tree/56fc236/studies/2026-10-19-og-gyrated-octet (research branch, commit 56fc236; also attached to this record)
The idea that the Ogdoecontahedron lives in the gyrated octet as well as in the octet, the step to its family (OCD Stella, golden envelope, DICTO Jewel), the belted solid, the octahedrally truncated solid, cutting the triacontahedron at more of its corners, and filling the space the solids leave are DICTO's; he chose the names of the solids and of the fill sets. Computation, checking and drafting with Claude (Anthropic).

DRAFT, Version 5, for DICTO's reading. Not published. Delete this box at publication. Version 5 adds a pointer to Pentagons for the Kites (#95): one sentence at the end of §3 and reference [13]; nothing else changed.



Abstract. In a companion paper the DICTO Ogdoecontahedron (OG) sits in every octahedron of the octet truss, its eight joint hexagons on the eight faces. DICTO proposed that it can also live in the *gyrated* octet, the tetrahedral–octahedral honeycomb stacked A B A B, in which octahedron meets octahedron and tetrahedron meets tetrahedron across every layer. It can: the OGs on both sides of a layer meet on the same joint hexagon, but each is the mirror image of the other, and since the OG has no such mirror the layers are **twin boundaries**; exactly, the twin is the OG turned 60° about the joint three-fold axis, the spinel law of crystallography. Cutting each octahedron into the OG and eight chiral caps, and keeping the tetrahedra whole, gives **one exact kit for the octet, the gyrated octet and every stacking between**; at a twin layer a left and a right cap join into a twin cap and two tetrahedra into a triangular bipyramid. The facing-star middle piece of the octet fails at twin layers, and a mirror middle piece of volume $(424 - 88\sqrt{5})/4617$ takes its place; with one boundary cap and one boundary tetrahedron the octet's Penrose-core kit then fills the gyrated octet and a single twin layer exactly, and with the same pieces any stacking; its Penrose cores lie only on the edges within the layers. The trait runs in the family: the OCD Stella twins in the same way, meeting its twin on a whole joint triangle; the golden envelope does not fit as it is, but cut flat at two opposite three-fold corners it becomes a solid of its own, the **DICTO Belted Triacontahedron (DBT)**, which fills the gyrated octet's octahedra pole to pole without overlap, 55.02% of space. Cut the same way at all eight of the corners that point through the octahedron's faces, it becomes the **DICTO Octahedrally Truncated Triacontahedron (DICTO OTT)**, with the OG's own symmetry, which twins like the OG, lies inside its octahedron and so fits every stacking, 52.89% of space. For each, the rest of space is filled by one more shape (two hands of one in the octet): the **DICTO Fill sets**. The DICTO Jewel's packing, framed by cubes, does not continue across a twin layer. All claims are exact in $\mathbb{Q}(\sqrt{5})$ except where marked float.

1. The gyrated octet

The octet truss (Fuller [1]) is the tetrahedral–octahedral honeycomb: regular octahedra and tetrahedra, two tetrahedra per octahedron, every octahedron face against a tetrahedron. Cut perpendicular to a three-fold axis it falls into layers one octahedron thick, each layer filling the slab between two parallel planes [2, 3]. Stacking the layers so that each is the mirror image of the one below gives the **gyrated** honeycomb, the hexagonal-close-packed (A B A B) counterpart of the face-centred-cubic (A B C) octet; any sequence of the two stackings tiles space as well, periodic or not [2, 4]. In the gyrated octet every layer boundary is a mirror plane: across it an octahedron meets an octahedron face to face, and a tetrahedron meets a tetrahedron, the pair forming a triangular bipyramid.

Frame (as in [5]). The OG at the origin, its octahedron $|x| + |y| + |z| \le R$ with $\mathbf{R = 2\varphi/3}$, volume $(64 + 32\sqrt{5})/81$; tetrahedra $(16 + 8\sqrt{5})/81$; a cell of the honeycomb (one octahedron and two tetrahedra) $(32 + 16\sqrt{5})/27$. A twin layer is the plane $x + y + z = R$ and its parallels. All coordinates are pairs of rationals $a + b\sqrt{5}$; corners are solved and checked exactly, volumes are exact sums of determinants.

2. The OG at a twin layer (exact)

The same hexagon. The OG's eight joint hexagons lie on the eight faces of its octahedron [5]. Reflected in the face $x + y + z = R$, the OG gives the OG of the octahedron on the other

side, and the two meet in the same six corners: the regular joint hexagon, side² = 2/9 − 2√5/27, covering (7 − 3√5)/2 ≈ 14.59% of the face.

A true twin. The reflected OG is not a translate of the OG (as it is in the octet and in the OG's body-centred cubic packing), because the OG, a pyritohedral solid, has no mirror plane perpendicular to a three-fold axis. Exactly, **the twin is the OG turned 60° about the joint three-fold axis**, checked with the rational rotation that turns (1,0,0) to (2/3, 2/3, −1/3). This is the spinel law of cubic crystals [6], whose twin plane is parallel to an octahedron face and whose twin operation is a half-turn about the ⟨111⟩ axis: because the OG has the 120° turn about that axis, the half-turn and the 60° turn give the same twin. Round the hexagon each edge opens into a wedge between the twins: a face beside the hexagon and its mirror image make an angle whose cosine is exactly (7 − 36√5)/109, an opening of about 47.6°.

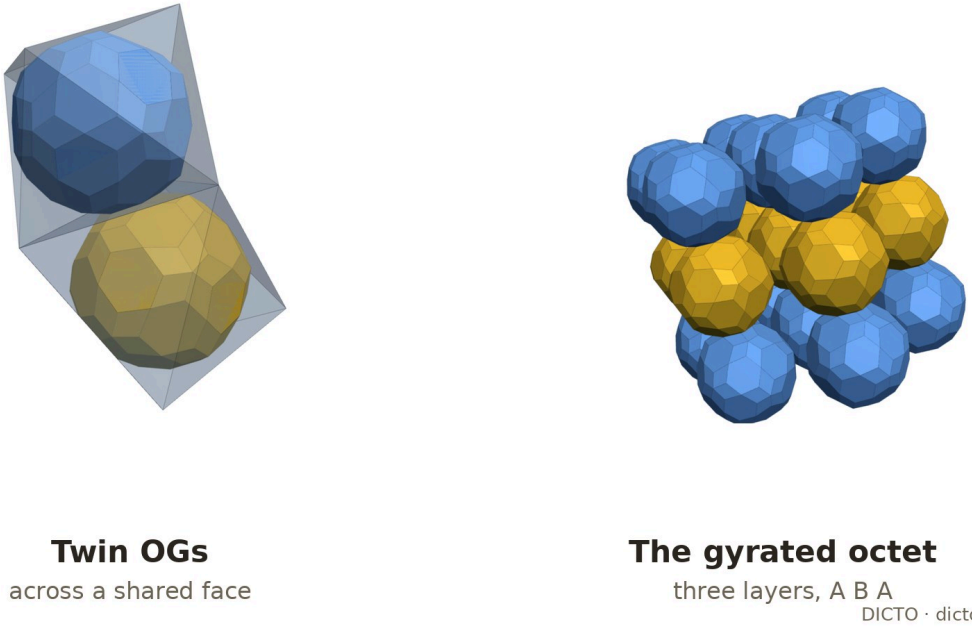


Figure 1. Left: the OG (gold) and its twin (blue) in two octahedra sharing a face; they meet on the same joint hexagon, and the twin is the OG turned 60°. Right: OGs in three layers of the gyrated octet; the layers above and below are the mirror image of the middle one.

3. One cell kit for every stacking (exact)

Cut each octahedron into the OG and **8 caps**, one under each face, by the planes through the centre and the octahedron's edges; keep each tetrahedron whole. Each cap is (octahedron − OG)/8 = **(277 − 79√5)/1539 ≈ 0.065205**, and the OG is (−1000 + 1240√5)/1539 ≈ 1.151868, (1025 − 435√5)/76 ≈ 68.83% of its octahedron. No piece crosses a face of its cell, so **the same kit fills the octet, the gyrated octet and any stacking of the layers**, at the same density.

The caps are chiral. The OG's three-fold turn about a face's axis maps the 18 OG corners in that face's cone onto themselves, but none of the three mirrors through the axis does. So the caps come in two hands, 4 + 4 per OG: the faces (1,1,1), (1,−1,−1), (−1,1,−1), (−1,−1,1) carry one hand, the other four the other (inversion swaps them).

At a twin layer a cap meets its own mirror image, a cap of the other hand, across the part of the face outside the joint hexagon, $(3\sqrt{5} - 5)/2 \approx 85.41\%$ of the face. Fused, they make a mirror-symmetric **twin cap** of $(554 - 158\sqrt{5})/1539 \approx 0.130410$. The two tetrahedra that meet across the layer make a **triangular bipyramid** of $(32 + 16\sqrt{5})/81 \approx 0.836754$. A gyrated octet can therefore be built from the OG, the left and right caps, the twin cap and the bipyramid (and the tetrahedron, where layers stack as in the octet). For the cut OG, the OG with its twelve five-fold tips cut into pentagons, kits for a single twin layer, the fully gyrated octet and mixed stackings, seven shapes for every stacking, are given in Pentagons for the Kites [13].

4. The facing-star middle piece at a twin layer

In the octet each octahedron edge joins two OGs whose five-fold corners face each other exactly (axes 180° apart, 0.3629 apart), and a middle piece spans them, with a thick Penrose rhombus between the stars [5, §3]. In the gyrated octet only the edges lying in a twin layer change: round them the cells run octahedron, tetrahedron | tetrahedron, octahedron, so the OG across the edge is the mirror twin, not the octet's partner. Middle pieces off the layer stay wholly on their own side (float check: no volume crosses the plane).

- **The octet's middle piece fails there** (float): about half of it (0.0399 of 0.0797) lies beyond the layer, 0.0198 of that inside the twin OG.
- **The stars no longer face** (float): the twin's five-fold corner is 0.1407 from the first, its axis turned 105.25° from it; the thick-rhombus link of the octet has no counterpart.
- **A mirror middle piece fits** (exact volume): the hull of a five-kite star and its mirror image, $(2 + 2\sqrt{5})/81$, minus the two OGs, each of which it meets in $(-155 + 101\sqrt{5})/4617$, gives **$\mathrm{Mg} = (424 - 88\sqrt{5})/4617 \approx 0.049215$**, one solid, mirror-symmetric (so with no left and right). In a float check against 43 neighbouring pieces and every OG it overlaps nothing.

The boundary pieces (exact). In the octet's keyless Penrose-core kit [5, §3] each octahedron is cut into the OG and six corner caps, every edge carries a middle piece (the hull of the two facing stars less both OGs), and caps and tetrahedra are notched by the middle pieces. At a twin layer the middle piece on each edge of the layer is Mg, and only two more shapes change: a **boundary cap**, a cap whose corner lies on a twin layer, $23/162 - 47\sqrt{5}/1215 \approx 0.055477$ (the octet's cap is $(32 - 8\sqrt{5})/243 \approx 0.058072$), and a **boundary tetrahedron**, a tetrahedron with a face on a twin layer, $41/162 + 4\sqrt{5}/135 \approx 0.319340$ (the octet's keyless tetrahedron $(26 - 2\sqrt{5})/81 \approx 0.265776$), in left and right hands as before. Computed by convex pieces for three stackings round one layer, the octet, a single twin layer and the gyrated octet:

- the octet run gives the octet's values [5]; with a single twin layer exactly the three caps and the three tetrahedra on that layer are boundary pieces, and the others octet pieces; in the gyrated octet every cap and every tetrahedron is one;
- no two middle pieces round the central octahedron overlap, and none enters a third OG;
- the octahedron (OG + 6 caps + the middle pieces' parts inside it) and a cell of the gyrated octet (OG + 6 boundary caps + 2 boundary tetrahedra + 3 middle pieces + 3 Mg = $(32 + 16\sqrt{5})/27$) sum exactly.

So **the octet's Penrose-core kit fills every stacking with three added shapes**: Mg, the boundary cap and the boundary tetrahedron (each piece feels only the layer it touches). The

Penrose cores stay on the middle pieces of the edges within a layer, three per cell in the gyrated octet against six in the octet; Mg has none, since the stars across a twin layer do not face. That each boundary piece is one solid and the same shape at every site is a float check.

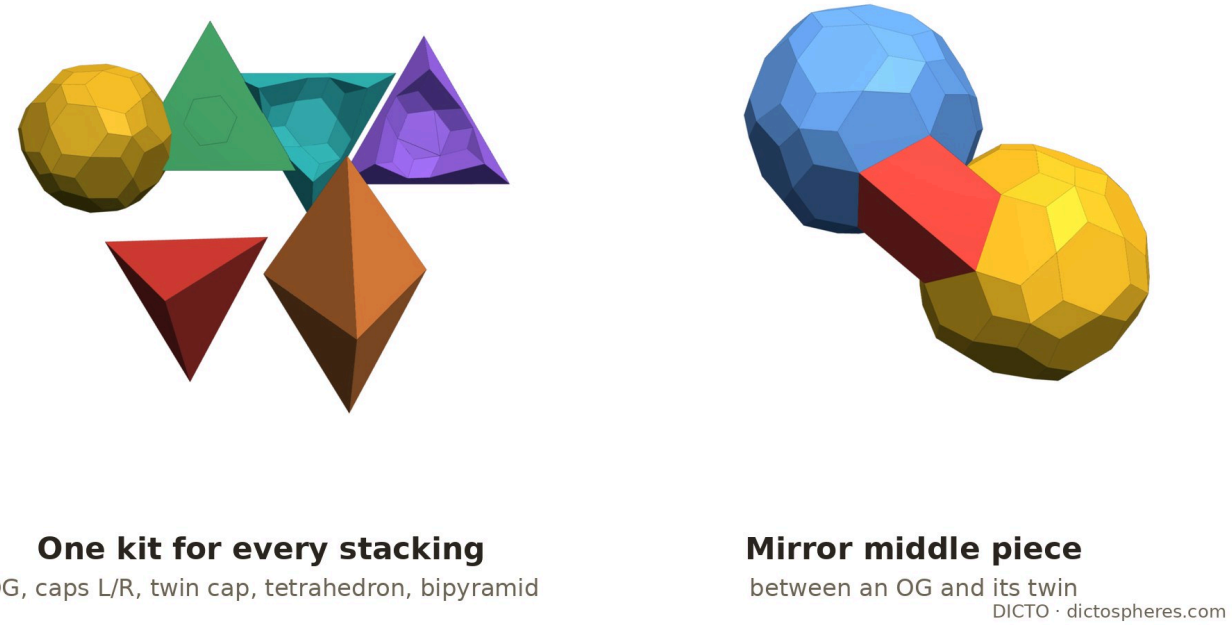


Figure 2. Left: the cell kit: the OG, the left and right caps (green, teal; the right cap shown from inside, with the OG's imprint), the twin cap (violet), the tetrahedron and the triangular bipyramid. Caps shown face-on. Right: the mirror middle piece Mg (red) between an OG (gold) and its twin (blue) at an edge of a twin layer.

5. The family (exact)

Four of the OG's relatives are tested: the OCD Stella; the golden envelope, which as it is does not fit, but cut at two of its corners gives the DBT (§6) and at eight the OTT (§7), both of which twin; and the DICTO Jewel, whose packing does not.

The OCD Stella (DICTO DOI: 10.5281/zenodo.23430870), the stella cut of the OG's golden companion in its body-centred cubic packing, has its eight joint faces on the same eight planes. Turned into this frame by the cube axes of that packing (the turn carries the packing's OG exactly onto the OG here), it lies in the octahedron and fills $(604 - 267\sqrt{5})/9 \approx 77.44\%$ of it. Each joint face is an equilateral triangle of side^2 $8/3 - 8\sqrt{5}/9$, $(7 - 3\sqrt{5}) \approx 29.18\%$ of the octahedron face: exactly twice the OG's joint hexagon. Its mirror twin in a face is again the Stella turned 60°, and **twin Stellas meet on the whole joint triangle**.

The golden envelope (the rhombic triacontahedron of inradius 2/3 round the OG) does not fit as it is (float): its three-fold corners poke 0.0909 beyond the faces, and across a twin layer two envelopes would overlap by about 0.0175 per shared face. In the octet those corners become keys [5, §2]; at a twin layer they have nowhere to go. Cut off at the two faces on one axis they give the DBT (§6); cut off at all eight, the OTT (§7).

The DICTO Jewel (first released with the Kaleidohedra app, DICTO DOI: 10.5281/zenodo.23223555): its packing does not twin (float, ray-parity sampling): in its Stella–Jewel lattice a (111) twin through a Jewel layer leaves 4.0% of the sampled points near the layer uncovered and 4.2% covered twice; through a stella layer 5.1% and 5.9%; between

layers 18–19%. The Jewel and the stella octangula are framed by cubes, and a cube mirrored across a plane at right angles to its body diagonal is the cube turned 60°, which no longer lines up with the cube lattice, whereas an octahedron mirrored in its face is simply the next octahedron of the gyrated octet. As a solid the Jewel twins like the OG: it is centre-symmetric with the three-fold axes, so its mirror image is the Jewel turned 60° (float check; the general rule is in The DICTO OG Family on Every Stacking, #40). It is the packing, not the solid, that fails.

6. The DICTO Belted Triacontahedron (exact)

Cut the golden envelope flat by the two octahedron faces $x + y + z = \pm R$ on one three-fold axis. DICTO named the result the **DICTO Belted Triacontahedron (DBT)**; once cut it is a solid of its own, not a container.

- **No new corners.** Each cut passes exactly through the pole corner's three nearest corners, so the DBT keeps 30 of the triacontahedron's 32 corners and loses only the two poles. It has 60 edges and 32 faces: 24 golden rhombi, 6 half-rhombi and 2 equilateral pole triangles of side² $8/3 - 8\sqrt{5}/9$ (side ≈ 0.824045, the same as the Stella's joint triangle).
- **Along the long diagonals.** Each cut runs along a rhombus's long diagonal (diagonal ratio² = φ^2).
- **Volume** $(-872 + 440\sqrt{5})/81 \approx 1.381110$; each removed corner $(-44 + 20\sqrt{5})/81 \approx 0.008906$, exactly **1/6 of the oblate golden rhombohedron** on the pole's three edges, $(-88 + 40\sqrt{5})/27$.
- **Symmetry $\mathbf{D_{3d}}$** (order 12): the three-fold turn, inversion and the half-turns about the three two-fold axes perpendicular to the cut axis. Sphericity 0.959 (float).
- **It holds the OG**, 36 of whose corners touch its surface.

In the gyrated octet. Every layer boundary is a twin plane, so each octahedron has its two pole faces on twin planes, exactly where the DBT is flat. Its mirror twin in a pole face is the DBT turned 60° (a true twin, as the OG). With a DBT in every octahedron, layer by translation and the neighbouring layers as mirror images, **no DBT overlaps another**: exact separating planes for all 56 neighbours checked (18 in the same layer, 19 above, 19 below). Twin DBTs meet flat, pole triangle to pole triangle, and the six keys round the belt reach into the tetrahedra of their own layer as in the octet. The DBTs fill $(493 - 219\sqrt{5})/6 \approx 55.02\%$ of space (82.53% of each octahedron).

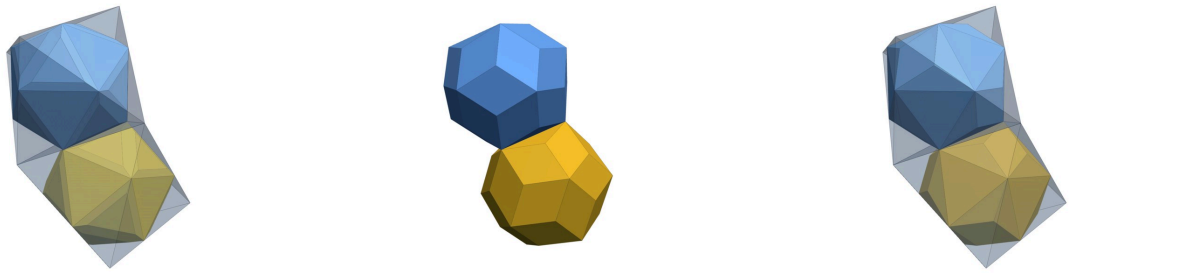


Figure 3. Left: an OCD Stella (gold) and its twin (blue) meeting on the whole joint triangle. Middle: two DICTO Belted Triacontahedra (gold and blue) pole to pole across a twin layer. Right: two DICTO Octahedrally Truncated Triacontahedra (gold and blue) meeting on the whole face triangle.

7. The DICTO Octahedrally Truncated Triacontahedron (exact)

The triacontahedron has twenty identical places for the DBT's cut, its twenty three-fold corners; at each the cut runs along three long diagonals and takes off the same golden pyramid, $(-44 + 20\sqrt{5})/81$. Eight of these corners point through the faces of the octahedron (they are the octet's keys [5, §2]); the DBT cuts two. DICTO proposed cutting more of them. **Cut at all eight**, the triacontahedron becomes its intersection with the octahedron, which DICTO named the **DICTO Octahedrally Truncated Triacontahedron (DICTO OTT)**.

- **No new corners.** 24 corners, all corners of the triacontahedron (the twelve five-fold corners lie exactly on the octahedron's twelve edges); 60 edges and 38 faces: 6 whole golden rhombi, 24 half-rhombi and 8 equilateral triangles of side^2 $8/3 - 8\sqrt{5}/9$, one on each octahedron face (the DBT's pole triangle).
- **Volume** $(-608 + 320\sqrt{5})/81 \approx 1.327676$, the triacontahedron less eight corners, or the DBT less its six keys.
- **Symmetry $\mathbf{T_h}$**, the OG's own pyritohedral symmetry, and not the full octahedral group. The OG lies inside it.
- **A true twin.** Its mirror image in an octahedron face is the OTT turned 60° about the face's axis, not a translate, as for the OG and the DBT; twin OTTs meet on the whole face triangle, which the mirror leaves in place. The triangle has the same corners as the OCD Stella's joint triangle on that face (float check).
- **Every stacking.** It lies in its octahedron, so the OTTs of the octet, of the gyrated octet and of any stacking never overlap; the tetrahedra stay whole. It fills $176/3 - 26\sqrt{5} \approx 52.89\%$ of space, $(-608 + 320\sqrt{5})/(64 + 32\sqrt{5}) \approx 79.33\%$ of its octahedron; anything inside the octahedra fills at most two thirds of space.
- **Size** (float). Scaling the triacontahedron by s before the cut: at $s = \varphi^2/3 \approx 0.873$ the eight corners just touch the faces; $s = 1$ is the golden fit above; for $1 < s < \varphi^2/\sqrt{5} \approx 1.171$ the cuts make 48 new corners; at $\varphi^2/\sqrt{5}$ the other twelve three-fold corners reach the faces (62.44%), and from there the solid is the octahedron with its corners cut by rhombus planes, until at $s = \varphi$ it is the octahedron itself. Cutting all twenty three-fold corners leaves the icosahedron of the twelve five-fold corners.

Packed by themselves (float; DICTO proposed checking how densely they pack alone). The octet and the gyrated octet place the solids for their golden joins and twin layers, not for density. Alone, both solids are centrally symmetric, and a search over lattices (copies at the points of one lattice, every lattice vector outside twice the solid, checked with the 26 nearest copies) finds lattice packings of density **0.906282 for the OTT** and **0.828621 for the DBT**; the same search gives 0.947368 = 18/19 for the octahedron (Minkowski) and 0.8018 for the rhombic triacontahedron (0.801741 in [12]). These are lower bounds for the densest packings; packings that are not lattices could be denser.

8. Filling the rest: the DICTO Fill sets (exact)

DICTO proposed filling the space the solids leave. As for the golden envelope and its grown piece in the octet [5, §4], every point outside the solids is given to its nearest site; a fill piece is the Voronoi region of a site less the solids. In the gyrated octet the sites are the bipyramids' centres (the centres of the tetrahedron faces on the twin layers, one per cell); in the octet they are the tetrahedron centres (two per cell), whose regions are cubes of edge R.

Set (DICTO's names)	Lattice	Piece	Volume	Hand	Sphericity (float)
DICTO Fill set 1 (DBT)	gyrated	F1, grown bipyramid	$(968 - 392\sqrt{5})/81 \approx 1.129153$	one	0.586
DICTO Fill set 2 (OTT)	gyrated	F1, grown bipyramid	$(704 - 272\sqrt{5})/81 \approx 1.182587$	one	0.612
DICTO Fill set 2 (OTT)	octet	F2, F3, grown tetrahedra	$(352 - 136\sqrt{5})/81 \approx 0.591293$	left, right	0.638

So each solid with one more shape fills space: the DBT and Fill set 1 F1 the gyrated octet; the OTT and Fill set 2 F1 the gyrated octet, and the OTT with F2 and F3 the octet. Exact, for every fill piece touching the central solid (6, 6 and 8): its volume, the Voronoi region less the solids in it; and an isometry of the whole local arrangement (solids and sites) taking the first piece to it, so each set has one shape throughout. A mirror of the arrangement fixes each bipyramid site, so F1 is its own mirror image; in the octet four of the eight pieces are reached only by mirrors and no mirror fixes a site, so F2 and F3 are left and right (a float check finds no mirror symmetry of the piece's own). That each piece is one solid is a float check. The cuts between fill pieces are flat Voronoi planes; the golden faces are the solids'.



Figure 4. Each solid with the fill pieces that touch it, pulled apart: the DBT with six Fill set 1 F1 (gyrated octet); the OTT with six Fill set 2 F1 (gyrated octet); the OTT with Fill set 2 F2 (green) and F3 (teal) in the octet.

9. Relation to known work, and what is claimed

The gyrated tetrahedral–octahedral honeycomb, its hcp stacking and the non-periodic stackings of the same layers are classical [2, 3, 4]; binary hard tetrahedra and octahedra self-assemble into the octet arrangement [7]. Twinning on an octahedron-face plane by a half-turn about ⟨111⟩ is the spinel law [6]; the gyrated honeycomb is described in [4] as reflective layers rotated 60°. Kabai's golden octet truss [8] is a module of one tetrahedral and one octahedral part of a golden rhombohedron, laid in planar and icosahedral lattices; his 54-faced space-filler [9] removes eight corners of a rhombic triacontahedron, each half an oblate golden rhombohedron, at the corners of a cube, for a body-centred cubic packing: a different cut and a different packing from the DBT, which removes two opposite corners, each one sixth of an oblate golden rhombohedron, and stacks as twins, and from the OTT, which removes eight, each one sixth, by the octahedron's own faces. The truncated rhombic triacontahedra described elsewhere cut the twelve five-fold corners (42 faces, full icosahedral symmetry) [10]; a catalogue of convex polyhedra with golden rhombi lists reduced and expanded triacontahedra only [11]. Giving the space outside golden solids to nearest sites is the method of the companion paper's grown piece [5, §4].

Not found in our reference library (the full text of the 80 sources in the Appendix, searched 20 October 2026) or in web searches the same day: icosahedral solids placed in the octahedra of the gyrated octet; the OG, the OCD Stella or the OTT twinned across its octahedron faces; a cell kit common to the octet and the gyrated octet; the twin cap, the mirror middle piece, the boundary cap and tetrahedron, the DBT or the OTT as a solid, or the fill sets. What is claimed is these results and their exact values; the honeycombs, the stackings and the twin law are not ours.

Open. Exact proofs that each boundary piece and each fill piece is one solid; the OTT and the fill sets in the Kaleidohedra app (the OG, OG + caps, OCD Stella and DBT on the gyrated octet are there, in The OG on the Octet Truss); physical prints of the kits.

10. Reproducibility

Exact scripts (ℚ(√5)): check.py (the shared hexagon, mirror ≠ translate, the wedge), twin.py (the 60° twin), kit.py and hand.py (caps, chirality), gyr_exact.py (hexagon share, twin cap, bipyramid, Mg, the DBT in the gyrated octet), stella_twin_exact.py (the OCD Stella), dbt_exact.py (the DBT), gyr_boundary_exact.py (the Penrose-core boundary pieces), cut8_exact.py and ott_twin_exact.py (the OTT), fill_exact.py (the fill sets). Float: gyr_cores.py (middle pieces at a twin layer), belted.py, jewel_twin.py and jewel_twin_planes.py (the Jewel), gyr_boundary.py, dbt_grow.py, cut8_size.py, dbt_fill.py, fill_hands.py, kits_page_data.py, lattice_pack.py. Logs: dbt-exact.log, gyr-exact.log, stella-twin.log, gyr-cores.log, gyr-boundary-exact.log, cut8-exact.log, ott-twin-exact.log, cut8-size.log, fill-exact.log, fill-hands.log, lattice-pack.log. Figures: gyr_page.py (meshes, float copies of the exact solids), shoot.mjs, make_figs.py, cover_lineup.py, cover_shots.mjs, cover_compose.py, fig_clusters.py, fig_clusters_compose.py. All at the commit given above.

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8. S. Kabai, Golden Octet Truss, Wolfram Demonstrations Project (2007).
9. S. Kabai, 54-Faced Space-Filling Polyhedron, Wolfram Demonstrations Project (2007).
10. Truncated Rhombic Triacontahedron, Wolfram Demonstrations Project, https://demonstrations.wolfram.com/TruncatedRhombicTriacontahedron/ (accessed 20 October 2026).
11. Convex Polyhedra containing Golden Rhombi, http://www.orchidpalms.com/polyhedra/rhombic/RTC/RTC.htm (accessed 20 October 2026).
12. S. Torquato, Y. Jiao, Organizing principles for dense packings of nonspherical hard particles: not all shapes are created equal, Phys. Rev. E 86, 011102 (2012), arXiv:1206.3716 (Table II).
13. DICTO, Pentagons for the Kites, #95, DICTO DOI: 10.5281/zenodo.23968270.

Appendix. The reference library searched

DICTO's reference library (private copies, with public citations in the Kaleidohedra repository's docs/SOURCES.md) holds 263 entries. For this paper three were added on 20 October 2026: Cadotte et al. [7] (arXiv copy) and the Wikipedia pages [4] and [6] (CC BY-SA). The full text of the 80 sources below was searched on 20 October 2026 for this paper's subjects: the gyrated honeycomb; hcp and stacking; twins and the spinel law; the tetrahedral–octahedral honeycomb and octet truss; bipyramids; the rhombic triacontahedron and its truncations; golden rhombohedra; pyritohedral symmetry; mirror images and enantiomorphs; and the names Ogdoecontahedron, OG and belted. None contains the results of this paper. The relevant hits are the honeycomb and its stackings [2, 3, 4, 7], the spinel law [6], and Kabai's golden octet truss and 54-faced space-filler [8, 9]; the word “gyrated” also occurs in a virology paper (Twarock and Luque 2019) for a two-dimensional lattice, unrelated. Thirteen further library items are page images without a text layer and could not be searched by text: the patents of Fuller (1954, 1961, 1962), Snelson (1965), Pearce (1971, 1976), Baer (1973), Hogan (1976), Bergman (1988) and Lalvani (1988, 1995, 1997), and Fedorov (1885); their subjects are recorded in the library index.

Web searches (20 October 2026): polyhedra inscribed in the octahedra of the gyrated tetrahedral–octahedral honeycomb, and twins in hcp stackings; the spinel law as a 60° turn or a (111) mirror; icosahedral space-filling solids in the octet or at its twin boundaries with the golden ratio; and the bibliographic details of [7].

Author(s)	Year	Title
Luca Pacioli, Leonardo da Vinci (ill.)	1509	Divina proportione
Albrecht Dürer	1525	Underweysung der Messung mit dem Zirckel und Richtscheyt
Wenzel Jamnitzer, Jost Amman (engr.)	1568	Perspectiva corporum regularium
Johannes Kepler	1596	Prodromus dissertationum cosmographicarum continens Mysterium cosmographicum
Johannes Kepler	1619	Harmonices mundi libri V
Euclid, T. L. Heath (tr.)	1908	The Thirteen Books of Euclid's Elements, Vol. III (Books X-XIII)
M. Goldberg	1937	A class of multi-symmetric polyhedra
Haresh Lalvani	1990	Transformational part-count in layered octahedral-tetrahedral truss configurations (NASA TM-102635)
P. W. Messer	1995	Stellations of the rhombic triacontahedron and beyond
M. Baake, F. Gähler	1998	Symmetry structure of the Elser-Sloane quasicrystal
T. C. Hales	1998	An overview of the Kepler conjecture
M. Baake	1999	A guide to mathematical quasicrystals
O. Ogievetsky, Z. Papadopolos	1999	On Quasiperiodic Space Tilings, Inflation and Dehn Invariants
U. Betke, M. Henk	1999	Densest lattice packings of 3-polytopes
Z. Papadopolos, O. Ogievetsky	1999	On Inflation Rules for Mosseri-Sadoc Tilings
Z. Papadopolos, O. Ogievetsky	1999	On quasiperiodic space tilings, inflation and Dehn invariants
R. V. Moody	2000	Model sets: a survey
M. Senechal	2004	Donald and the golden rhombohedra
S. Kabai	2007	54-Faced Space-Filling Polyhedron (Wolfram Demonstration; description and screenshot)
S. Kabai	2007	Golden Octet Truss (Wolfram Demonstration; description and screenshot)
Wikipedia contributors (2007 mirror)	2007	Kyushindo (2007 version)
S. Kabai	2008	A Rhombic Dodecahedron Made of Rhombic Triacontahedra (Wolfram Demonstration; description and screenshot)
S. Kabai	2008	Relating a Rhombic Triacontahedron and a Rhombic Dodecahedron (Wolfram Demonstration; description and screenshot)
M. Koca, M. Al-Ajmi, N. O. Koca	2009	Grand antiprism and quaternions
M. Koca, M. Al-Ajmi, N. O. Koca	2009	Quaternionic representation of snub 24-cell and its dual polytope derived from E8 root system
S. Torquato, Y. Jiao	2009	Dense packings of polyhedra: Platonic and Archimedean solids
B. Grünbaum	2010	The Bilinski dodecahedron, and assorted parallelohedra, zonohedra, monohedra, isozonohedra and otherhedra
P. Kramer	2010	Gateways towards quasicrystals
A. M. Cutler, E. Schulte	2011	Regular polyhedra of index two, I
D. Frettlöh	2011	Icosahedral tilings in R3: the ABCK tilings (draft chapter)
J. Koskinen	2011	Unusual tilings and transformations
J. de Graaf, R. van Roij, M. Dijkstra	2011	Dense regular packings of irregular non-convex particles
M. Koca, N. O. Koca, M. Al-Ajmi	2011	Branching of the W(H4) polytopes and their dual polytopes under the Coxeter groups W(A4) and W(H3) represented by quaternions
M. Koca, N. O. Koca, M. Al-Barwani	2011	Snub 24-cell derived from the Coxeter-Weyl group W(D4)
T. Etzion	2011	Product constructions for perfect Lee codes
A. Garber, A. Gavrilyuk, A. Magazinov	2012	The Voronoi conjecture for parallelohedra with simply connected delta-surface
J. C. Lagarias, C. Zong	2012	Mysteries in packing regular tetrahedra
M. Baake, U. Grimm	2012	Mathematical diffraction of aperiodic structures
M. Koca, N. O. Koca, R. Koc	2012	Affine Coxeter group Wa(A4), quaternions, and decagonal quasicrystals
P. F. Damasceno, M. Engel, S. C. Glotzer	2012	Crystalline assemblies and densest packings of a family of truncated tetrahedra and the role of directional entropic forces
R. Gabbrielli, Y. Jiao, S. Torquato	2012	Families of tessellations of space by elementary polyhedra via retessellations of face-centered-cubic and related tilings
S. Torquato, Y. Jiao	2012	Organizing principles for dense packings of nonspherical hard particles: not all shapes are created equal
M. Kléman	2013	Defects in quasicrystals, revisited: I-flips, approximants, phason defects
N. Fujita, H. Takano, A. Yamamoto, A.-P. Tsai	2013	Cluster packing geometry for Al-based F-type icosahedral alloys
P.-P. Dechant, J. Wardman, T. Keef, R. Twarock	2014	Viruses and fullerenes: symmetry as a common thread?
A. Campello, G. C. Jorge, J. E. Strapasson, S. I. R. Costa	2015	Perfect codes in the lp metric

K. Deguchi, M. Nakayama, S. Matsukawa, K. Imura et al.	2015	Crystal structure of superconducting 1/1 cubic Au-Ge-Yb approximant with Tsai-type cluster
M. Baake, D. Damanik, U. Grimm	2015	What is aperiodic order?
M. Firsching	2015	Computing maximal copies of polytopes contained in a polytope
N. O. Koca, M. Koca, A. Al-Mukhaini, A. Al-Qanobi	2015	Quaternionic representations of the pyritohedral group, related polyhedra and lattices
R. Twarock, M. Valiunas, E. Zappa	2015	Orbits of crystallographic embedding of non-crystallographic groups and applications to virology
S. Kabai	2015	Cluster of 14 Rhombic Triacontahedra (Wolfram Demonstration; description and screenshot)
A. T. Cadotte, J. Dshemuchadse, P. F. Damasceno, R. S. Newman, S. C. Glotzer	2016	Self-assembly of a space-tessellating structure in the binary system of hard tetrahedra and octahedra
C. Hann, J. E. S. Socolar, P. J. Steinhardt	2016	Local growth of icosahedral quasicrystalline tilings
R. M. K. Dietl, J.-H. Eschenburg	2016	The icosahedral quasiperiodic tiling and its self-similarity
A. Stergiou	2018	Bootstrapping hypercubic and hypertetrahedral theories in three dimensions
A. Garber	2019	Voronoi conjecture for five-dimensional parallelohedra
D. Poland, S. Rychkov, A. Vichi	2019	The conformal bootstrap: theory, numerical techniques, and applications
R. Twarock, A. Luque	2019	Structural puzzles in virology solved with an overarching icosahedral design principle
T. Ishimasa	2019	Icosahedral quasicrystal, 1/1 and 2/1 approximants in Zn-based ternary alloys containing Au and Yb/Tb
A. Al-Siyabi, N. O. Koca, M. Koca	2020	Icosahedral polyhedra from D6 lattice and Danzer's ABCK tiling
M. Koca, R. Koc, N. O. Koca, A. Al-Siyabi	2020	Icosahedral tiling with dodecahedral structures
N. O. Koca, R. Koc, M. Koca, A. Al-Siyabi	2020	Dodecahedral Structures with Mosseri-Sadoc Tiles
V. R. Krishnamurthy, E. Akleman, S. G. Subramanian et al.	2020	Bi-axial woven tiles: interlocking space-filling shapes based on symmetries of bi-axial weaving patterns
V. R. Krishnamurthy, E. Akleman, S. G. Subramanian, M. Ebert, J. Cui, C. Fu, C. Starrett	2022	Geometrically interlocking space-filling tiling based on fabric weaves
B.-X. Lao, S. Rychkov	2023	3D Ising CFT and exact diagonalization on icosahedron: the power of conformal perturbation theory
D. Smith, J. S. Myers, C. S. Kaplan, C. Goodman-Strauss	2023	An aperiodic monotile
R. Akpanya et al.	2024	Topologically interlocking blocks inside the tetroctahedrille
I. J. Dejter, L. R. Fuentes, C. A. Araujo	2025	Perfect truncated-metric codes in lattice graphs and superlattice graphs
S. Rychkov	2025	Conformal bootstrap: from Polyakov to our times
M. Kovačević	2026	An isodiametric theorem and lattice diameter-perfect codes in A3
R. Al Raisi, N. O. Koca, M. Koca, R. Koc	2026	Modified Mosseri-Sadoc tiles from D6
Wikipedia contributors	2026	Crystal twinning (incl. the spinel law)
Wikipedia contributors	2026	Fullerene
Wikipedia contributors	2026	Goldberg polyhedron
Wikipedia contributors	2026	Kenshiro Abbe
Wikipedia contributors	2026	Kyūshindō
Wikipedia contributors	2026	Mitsusuke Harada
Wikipedia contributors	2026	n-flake (Sierpinski icosahedron)
Wikipedia contributors	2026	Tetrahedral-octahedral honeycomb (incl. the gyrated form)